

DATA ANALYTICS PRACTICE SERIES



Apple Sales Analytics

NumPy Workbook

Beginner to Advanced

Project-Based Data Analytics Practice

9 Blocks • 45 Questions • Portfolio Ready

TABLE OF CONTENTS

What's Inside

01	Array Fundamentals	4
02	Indexing & Slicing	5
03	Reshape, Stack & Split	6
04	Vectorized Operations & Broadcasting	7
05	Aggregation & Axis Operations	8
06	Boolean Masking & Filtering	9
07	Sorting, Unique & Set Operations	10
08	Statistics & Numerical Analysis	11
09	Final Business Challenges	12

NOTE

Each block progresses from foundational recall (Easy) through applied reasoning (Medium) to a Business Challenge (Hard) rooted in the Apple Sales dataset.

PART ONE

NumPy Workbook

Nine blocks. Forty-five questions. From array fundamentals to a production-ready feature matrix built on real Apple sales data.

BLOCK 01

Array Fundamentals

TOPICS COVERED

NumPy Arrays

ndarray

dtype

shape

size

ndim

Array Creation

Q1

EASY

Load `sales.csv` and convert the Quantity column into a NumPy array.

Q2

EASY

Display the shape, size, ndim, and dtype of the Quantity array.

Q3

MEDIUM

Convert the Quantity array from Integer to Float.

Q4

MEDIUM

Create a Zero Array having the same length as the Quantity array.

Q5

HARD – BUSINESS CHALLENGE

Create a 2D NumPy matrix using Sale ID, Store ID, Product ID, and Quantity. Inspect the matrix and explain why this structure is useful for business analytics.

BLOCK 02

Indexing & Slicing

TOPICS COVERED

Indexing

Slicing

Row Selection

Column Selection

Negative Indexing

Q1

EASY

Display the first 10 Quantity values.

Q2

EASY

Display the last 10 Quantity values.

Q3

MEDIUM

Display the middle 20 Quantity values.

Q4

MEDIUM

Extract Product ID and Quantity as a new NumPy matrix.

Q5

HARD – BUSINESS CHALLENGE

Retrieve the first 100 sales records using slicing and explain why slicing is preferred over loops for large datasets.

BLOCK 03

Reshape, Stack & Split

TOPICS COVERED

reshape()

flatten()

hstack()

vstack()

split()

Q1

EASY

Take the first 100 Quantity values and reshape them into a 10×10 matrix.

Q2

EASY

Flatten the reshaped matrix back into a 1D array.

Q3

MEDIUM

Horizontally stack Product ID and Quantity into a single matrix.

Q4

MEDIUM

Vertically stack two equal-sized Quantity arrays.

Q5

HARD – BUSINESS CHALLENGE

Split the first 1000 Quantity values into 10 equal batches for parallel processing.

BLOCK 04

Vectorized Operations & Broadcasting

TOPICS COVERED

Arithmetic Operations

Vectorization

Broadcasting

Q1

EASY

Increase every Quantity value by 1.

Q2

EASY

Multiply every Quantity value by 2.

Q3

MEDIUM

Simulate a festival sale by adding 10 bonus units to every sale using vectorized operations.

Q4

MEDIUM

Create a promotional multiplier array and apply it using broadcasting.

Q5

HARD – BUSINESS CHALLENGE

Simulate different stock replenishment values for different stores using broadcasting without writing loops.

BLOCK 05

Aggregation & Axis Operations

TOPICS COVERED

sum()

mean()

min()

max()

axis

Q1

EASY

Calculate the Total Quantity Sold.

Q2

EASY

Calculate the Average Quantity Sold.

Q3

MEDIUM

Find the Minimum and Maximum Quantity sold.

Q4

MEDIUM

Using a reshaped Quantity matrix, calculate row-wise and column-wise totals.

Q5

HARD – BUSINESS CHALLENGE

Build a Sales Summary Report containing Total Quantity, Average Quantity, Maximum Quantity, Minimum Quantity, and explain which KPI would be most useful for a business manager.

BLOCK 06

Boolean Masking & Filtering

TOPICS COVERED

Boolean Masking

Comparison Operators

Logical Operators

np.where()

Conditional Filtering

Q1

EASY

Display all sales where the Quantity is greater than 2.

Q2

EASY

Display all sales where the Quantity is equal to 1.

Q3

MEDIUM

Count the total number of sales where the Quantity is greater than the average Quantity.

Q4

MEDIUM

Replace every Quantity value greater than 5 with 5 using NumPy.

Q5

HARD – BUSINESS CHALLENGE

Create a filtered NumPy matrix containing only high-volume sales (Quantity > Average Quantity) and prepare it for further business analysis.

BLOCK 07

Sorting, Unique & Set Operations

TOPICS COVERED

sort()

argsort()

unique()

intersect1d()

setdiff1d()

Q1

EASY

Sort the Quantity array in ascending order.

Q2

EASY

Find all unique Quantity values present in the Sales dataset.

Q3

MEDIUM

Find all unique Product IDs that appear in the Sales dataset.

Q4

MEDIUM

Count the frequency of each Quantity value using NumPy.

Q5

HARD – BUSINESS CHALLENGE

Rank all sales based on Quantity using argsort() without modifying the original dataset and identify the Top 20 highest quantity sales.

BLOCK 08

Statistics & Numerical Analysis

TOPICS COVERED

mean()

median()

std()

var()

percentile()

Q1

EASY

Calculate the Mean Quantity sold.

Q2

EASY

Calculate the Median Quantity sold.

Q3

MEDIUM

Calculate the Variance and Standard Deviation of the Quantity values.

Q4

MEDIUM

Calculate the 25th, 50th, and 75th Percentiles of Quantity sold and interpret what they represent.

Q5

HARD – BUSINESS CHALLENGE

Normalize the Quantity values between 0 and 1 and identify potential outliers using statistical measures.

BLOCK 09

Final Business Challenges

TOPICS COVERED

Feature Engineering

Business Analysis

Numerical Analysis

Data Preparation

Q1

EASY

Identify all sales where the Quantity sold is above the overall average Quantity.

Q2

EASY

Display the Top 20 highest Quantity sales using NumPy only.

Q3

MEDIUM

Calculate the contribution percentage of every sale to the total Quantity sold.

Q4

MEDIUM

Simulate a 15% increase in product demand and calculate the updated total Quantity sold.

Q5

HARD – BUSINESS CHALLENGE

Create a final NumPy feature matrix containing Sale ID, Product ID, Store ID, Quantity, and Revenue (Quantity × Price). The matrix should be optimized and ready to be passed into the Pandas analysis phase.



NumPy Workbook Complete

- ✓ 9 Blocks
- ✓ 45 Questions
- ✓ Project-Based Learning
- ✓ Apple Sales Dataset
- ✓ Beginner → Advanced
- ✓ Portfolio Ready
- ✓ Interview Oriented